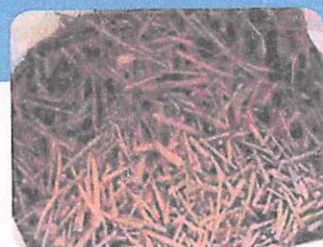




GASHARI NWFP MANAGEMENT PLAN, NORBUGANG, NGANGLAM (1ST REVISION)

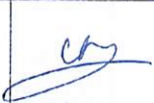
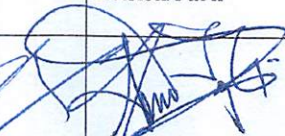
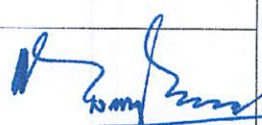
Plan Period: July 2021 – June 2026



APPROVAL SHEET

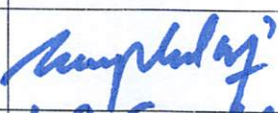
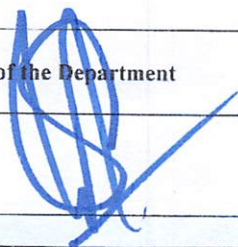
Name of NWFP management and Marketing Group:	Gashari NWFP Management and Marketing Group
Dzongkhag:	Pema Gatshel
Gewog:	Norbugang
Village(s):	Gashari
No. of Households:	9
Area(Ha):	64.3 Hectares
Plan Period:	2021 - 2026

Prepared and Submitted for Approval

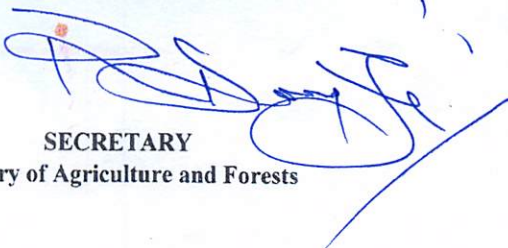
	Chairperson of NWFP Management and Marketing Group	Gup of the Gewog	Head Social Forestry Section of Division/Park	Chief Forestry Officer of Division/Park
Signature:				
Name:	Phuntsho Wangdi	Ugyen Dorji	Kinley Dorji	Ugyen Wangchuk
Date:	17/3/2021	26/3/21		

Chief Forestry Officer
Divisional Forest Office
Pema Gatshel

REVIEWED AND RECOMMENDED FOR APPROVAL

	Reviewing Officer, SFED	Chief Forestry Officer of the SFED	Head of the Department
Signature:			
Name:	SONAM PELDON	L. P. Tenzin	
Date:	24/06/2021	29/6/21	

APPROVED BY:


SECRETARY
Ministry of Agriculture and Forests

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GLOSSARY AND ABBREVIATIONS

CFD	Community Development Fund
CFO	Chief Forestry Officer
DoFPS	Department of Forests and Park Services
DT	Dzongkhag Tshogdue
GFEO	Gewog Forestry Extension Officer
GT	Gewog Tshogdue
HHs	Households
NTFPs	Non-timber Forest Products
NWFPs	Non-wood Forest Products
ToR	Terms of Reference

EXECUTIVE SUMMARY

Gashari Non- Wood Forest Products (NWFP) Management Group comprises of 9 members from Gashari community under Norbugang Gewog, Nganglam Dungkhag. The group was formed to manage four species of NWFPs which have potential for income generation. These resources will be collected from 13 identified sites designated for collection and management in this plan. The areas are Gasharibali, Takdzong, Lablagonpa, Garjoktang borang near RMNP, Yangkholom, Tanzema, Zirzila near CF areas, Chichari, LingshingGonglam, Tsenkaritoe, Wunbuzur, Tokaphung – Dotah, Zeyjab, and Gasharibali – Yangkholom areas.

The collection sites identified are dominated by NWFPs species such as *Rubia cordifolia*, *Thysanolaena latifolia*, *Piper pedicellatum*, and *Dendrocalamus spp.* The handing over of these collection sites to community will enhance sustainable management and utilization of resources. Further, it will enable the community to make income as there is a market for the products.

Management Plan consists of the brief description of NWFPs, descriptions of individual species including harvesting prescriptions, GIS map of NWFPs collection areas and marketing and sales plan. The by-laws of the management plan contain process of executive committee formation-ToR of Chairperson, Secretary and Treasurer, compensation to the executive committee for marketing, membership fee and community development fund, conflict resolution and penalties.

The term of management plan is for five years and will be effective from July 2021 to June 2026. During the period, they will have right to collect mentioned NWFPs as per the harvesting guidelines prescribed in the plan to ensure sustainability of resources in the future.

PART 1: BACKGROUND INFORMATION

1.1: Short descriptions of *Rubia* (*Rubia Cordifolia*)

Rubia (local name Laniru) is a perennial scrambling or climbing herbs or sub-shrubs of around 80 species native to various parts of the world. It can grow up-to 1.5 m in length. The evergreen leaves are 5-10cm long and 2-3cm broad, produced in whole whorls of 4-7 star like around the central stem. It climbs with tiny hooks at the leaves and stem. The flower usually appears from June to August. It grows along the forest edge and clearings,

scrub vegetation, dune forest, less commonly in grassland and open and rocky areas. It grows up to the elevation of 2600 masl.



Figure 1: *Rubia cordifolia* & its processes

Uses

Rubia is an economically important source of a red pigment in many regions of Asia, Europe and Africa. The plants roots contain an anthracene compound called alizarin hat gives its red color to a textile dye known as rose madder. It also used as a colourant especially for paints.

1.2: Short descriptions of *Pipla* (*Piper pedicellatum*)

Pipla Local name: (Bonzalo) is a perennial plant that grows as bushes to a height of 1-2 meter. It has smooth dark greenish leaves. Pipla is a dioceous plant, meaning that it has unisexual reproductive units with male and female plants separately. Only the female plants bear round berries that have commercial value. The fruits start ripening in August – September.

Pipla thrives widely under the canopy of broadleaf forests or in an open area within the altitudes ranges of 200-2200 masl in Zhemgang and Pemagatshel Dzongkhags, and in some other sub-tropical forests around the country. It is the shade demander and thrives mostly in damp and moist soil. There are eight

Pipla species found in different parts of Bhutan: *Piper pedicellatum*, *P. attenuatem*, *P. longum*, *P. rhytidocarpum*, *P. mullesua*, *P. peepuloides*, *P. sylvaticum* and *P. khasianum*. The bush type is more popular to the local farmers where the fruits have good commercial value and this management plan is only refers to *Piper pedicellatum* and cannot apply to other Pipla species.



Fig 2: *Piper pedicellatum*

Uses

The fruits have distinctive sour and hot taste that can be used for medicinal purposes, and also use as a species. In Indian Ayurvedic System as well as the Tibetan and Bhutanese Medical Tradition, recognize *Pipla* as a powerful stimulant for the digestive and respiratory system as a rejuvenating agent, longevity enhancer and tonic for the immune system

1.3: Short descriptions of Brooms Grass (*Thysanolaena latifolia*)

Thysanolaena latifolia (locally called Tsakusha) is popular for its rural livelihood ventures to many farmers in Bhutan. This species is found in Bhutan's sub-tropical areas. It grows as high as 2.5 meters and is mostly found in open areas particularly above the road stabilization site, erosion/landslide site, logged-over areas, mountain slopes and ravines. The leaves are linear, about 5 cm wide and 30 cm long; and are alternately arranged at an interval of 12.5 cm. The panicles which are the main material for soft-broom production has an average length of 50

cm while its stalk measures about 70 cm long. It can be grown in many types of soils such as clay, sandy clay loam and sandy loam. A one and one-half year old broom grass starts producing panicles that develop from October to March every year.



Fig 2: *Thysanolaena latifolia*

Uses

T. latifolia is the source of raw material used for making soft brooms. It is widely used as planting materials for controlling soil erosion and as agro forestry crop in many countries. It also used as live fencing and boundary plantations materials. The leaves of the plant can be good forage for animals like cattle and horses.

1.4: Shoot descriptions of bamboo (*Dendrocalamus spp.* and *Arundinaria spp.*)

Bhutan probably has the greatest variety of bamboo (**Local name:** Soo) species of all the Himalayan countries. Bhutanese bamboo is principally of Himalayan and Chinese-Japan origin with some Southeast Asian and South Indian contribution. There are 34 bamboo species recorded in Bhutan of which two species *Dendrocalamus* and *Arundinaria* are

popular in regions. So, the management plan focus only on two of these species.



Fig 4: Bamboo groves

Uses

Dendrocalamus spp.: Used as a roofs and wall of houses, domestic and agricultural implements such as water container, baskets, trays, mat, etc., livestock shelter and temporal dwellings, parts of traditional houses and edible shoots.

Arundinaria spp.: High-quality woven handcraft products such as food and drinks containers, hats, arrows and quivers, winter and dry season livestock fodder, durable mats for building constructions, fencing materials, edible shoots, shelter and food for endemic fauna in reserve areas.

PART 2: NWFPs MANAGEMENT AND MARKETING PLAN

2.1: The objectives of the management plan are given below:

- To enhance income generation to the community through sale of NWFPs.
- To promote and manage the NWFPs resources at local level through community participations.
- Poverty reduction and self sufficiency

2.2: NWFPs descriptions

Table 1: Block description of individual NWFPs species

NWFP name	Collection area (name)	Parts harvested	Use purpose (traditional and others)	Harvesting season (months)	Estimated quantity that can be collected
Laniru (<i>Rubia cordifolia</i>)	Gasharialey, Takdzong, Lablagonpa, Garjoktang boring near RMNP	Entire plants	Dye, medicine	November-February	200 kgs
Pipla (<i>Piper pedicellatum</i>)	Yangkholom, Gasharibali, Tanzema, Zirzila near CF areas	Fruits	Spice/ medicine young shoots are used as vegetable	Fruits are fully mature usually from Sept onwards	300 kgs
Bamboo (<i>Dendrocalamus spp.</i>)	Gasharibali, Chichari, LingshingGonglam, Tsenkaritoe, Wunbuzur	Culm, young shoot	Weaving, construction materials, etc.	For shoots: May to September Culms: Nov -March	150 nos.

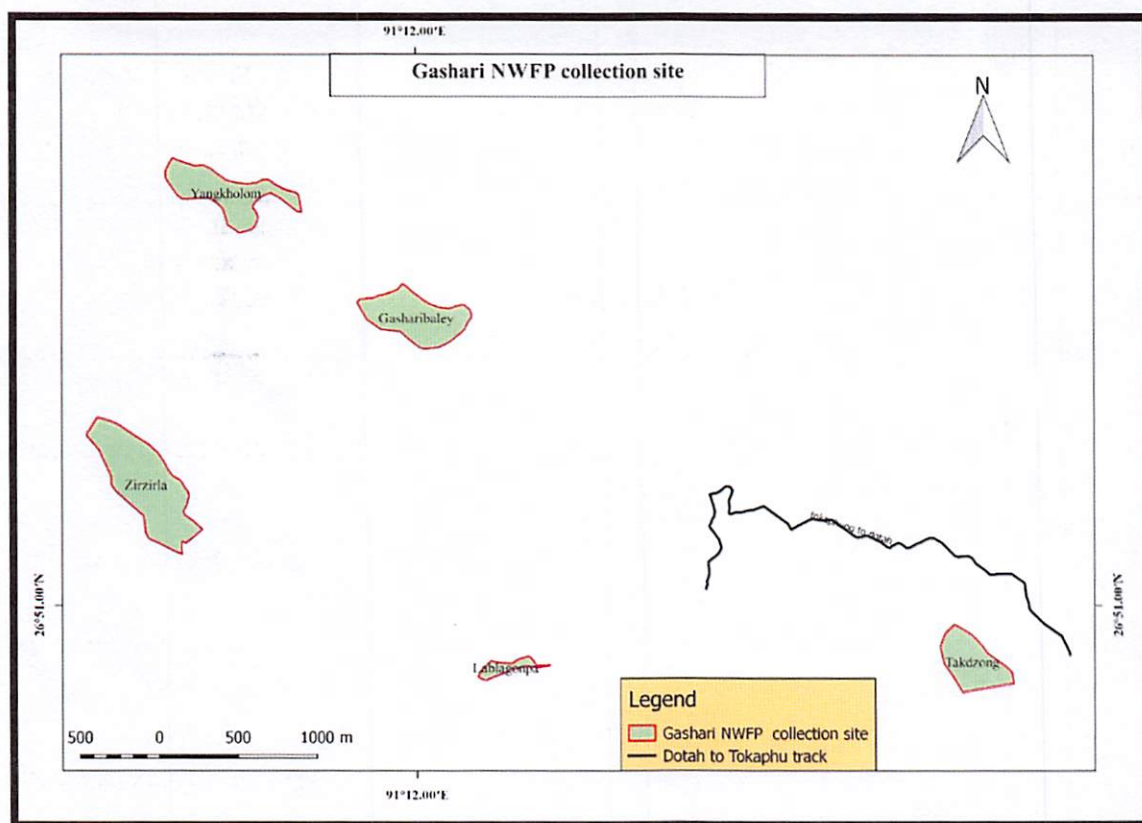
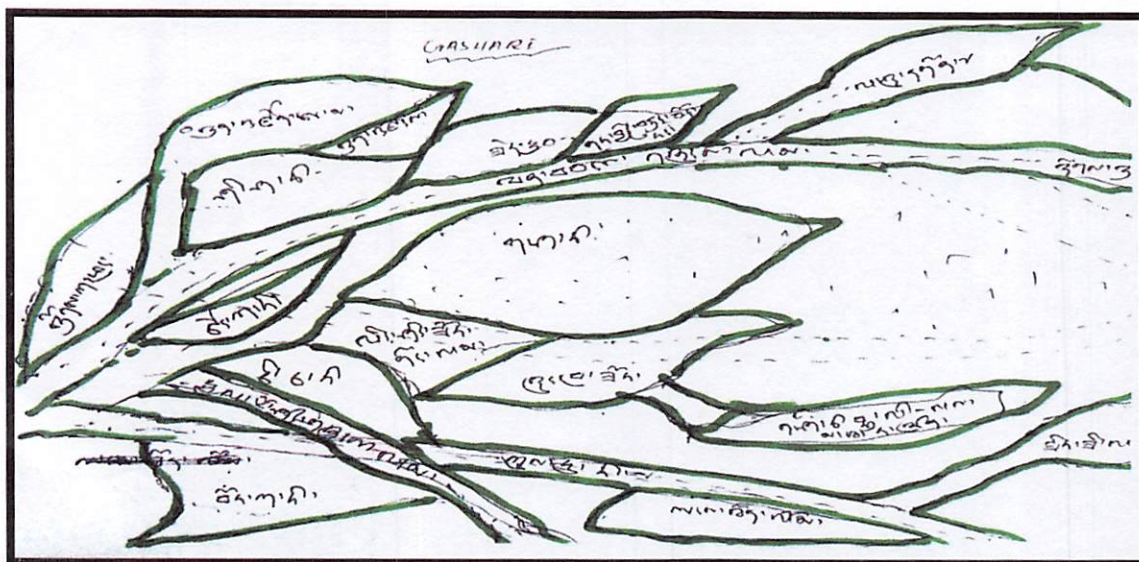
Tsakusha (<i>Thysanolaena latifolia</i>)	Tokaphung – Dotah, Zeyjab, Gasharibali - Yangkholom	Use upper parts, infloresce nce	Fodder, soft broom	December- March	300 bundles
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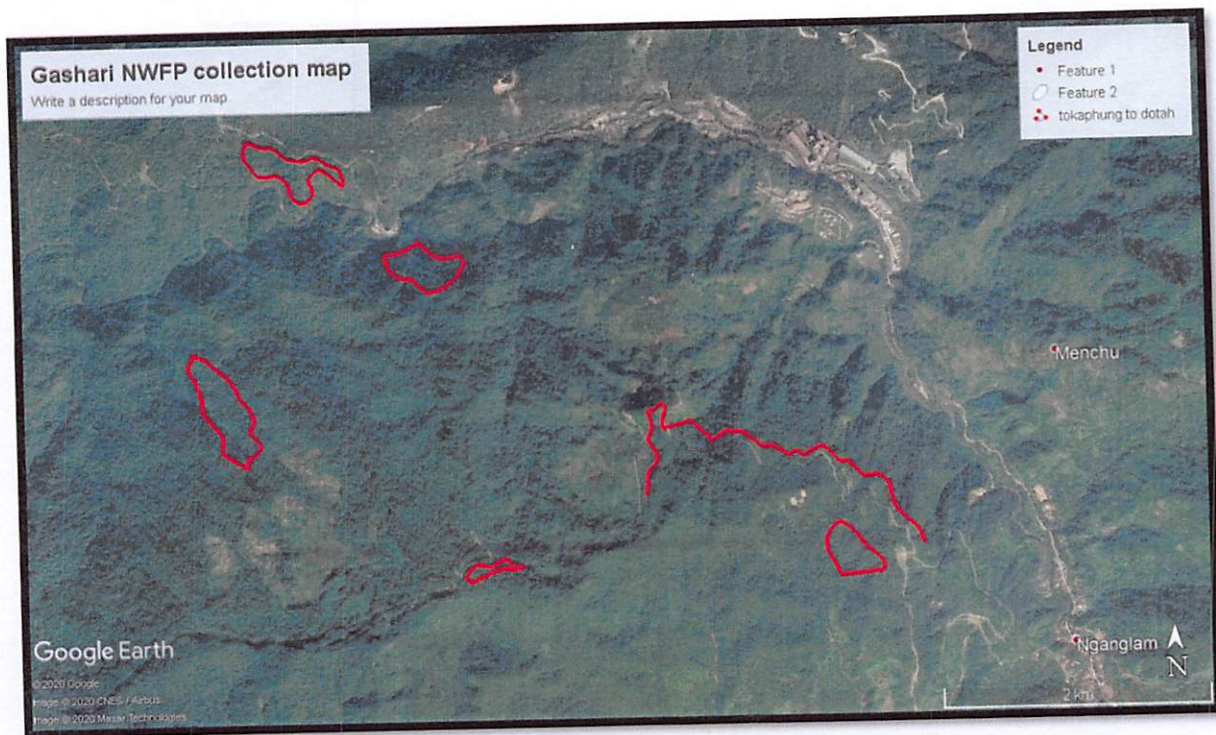
2.3: Harvesting prescriptions for sustainable NWFP harvesting

Table 2: Prescriptions for sustainable harvesting of NWFP

NWFP species	Harvesting prescriptions for sustainable harvesting
Laniroo (<i>Rubia cordifolia</i>)	-Use sharp sickle for the collection of the upper part of the plant. -Cut the plant at 5 cm above ground level. -Leave at least 10% of the plants without collecting or harvesting for regeneration. -Do not uproot roots and collect all plant for sale. -Do not uproot entire plants -Leave the mother plant intact with small part of the root or rhizome in the soil (this left part of rhizome will regenerate in the following year).
Bonzalo (<i>Piper pedicellatum</i>)	-Collect fruits only when fully matured. -Harvest fruits only after the announcement of start date for harvesting (Date will be announced by the coordinator/GFR generally in September) -Harvest fruits only by hand picking without breaking, cutting or felling of the entire plant
Tsakusha (<i>Thysanolaena latifolia</i>)	-Cut the foliage, inflorescence at 15 cm from the ground. -Do not damage the rhizomes in order to allow natural regeneration. -Do not put fire on rhizome to ease cleaning work.
Bamboos (<i>Dendrocalamus spp</i>)	For culms: -Cut culms above one node from the ground level to harvest matured culms. -Do not cut culms of less than one year old -Retain 50% of healthy culms in a clump -Harvest culms in condensed/congested clump in “U” or horseshoe form. Use otherwise single culm selection method -Harvest matured culms at center of clump first and proceeds outwardly. For shoots: -Collect bamboo shoots only to a maximum of 25% from a clump -Cut bamboo-shoots with a sharp knife at the ground level without damaging or wounding the other healthy ones -After cutting and collecting the bamboo-shoot, cover that void place with top soil. - Should not cut shoots of last flush. Shoots must be collected only from 1st and 2nd flush.

Figure 3: Map of the NWFPs collection area





2.5: Marketing and sales plan

Table 3: Marketing and sales plan of NWFPs

NWFP name	Potential market	Estimated quantity for sale per season	Expected price per unit	Total Nu.
Laniru (<i>Rubia cordifolia</i>)	Local trader, Indian market	200 kgs	80	16,000.00
Bonzalu (<i>Piper pedicellatum</i>)	Local trader, Indian market	300 kgs	200	60000.00
Tsakusha (<i>Thysanolaena latifolia</i>)	local market, trader, , Indian market	300 bundles	45	13500.00
Bamboo (<i>Dendrocalamus spp.</i>)	Local market, Local contractor, Indian market.	150 nos.	10.00	1500.00

2.6: Marketing arrangements

The members will collect the NWFPs individually and store properly. The members will grade products as per the standards required by the trader. The executive committee members will inform forestry official/DAMC and collect the NWFPs product to the auction yard. The concern forestry official/DAMC will then organize NWFPs auction with invitation of NWFPs traders. The auction will be organized in the presence of Chairman from the respective NWFPs groups and also interested members.

Upon completion of auction the executive members hand over products only after receipt of payment in hand. The community members will work out on deduction of expenditure on transportation, community fund and compensation. The balance amount will be shared individual group member based on the quantity of NFPs collected from respective members.

2.7: Benefits to the members

The individual members will have equal opportunities and access the resources. The members can earn income from the sale of NWFPs. This income will enable the members to meet expenses for purchase of basic necessities like salt, rice, sugar, and other supplementary food items. Furthermore, the income also helps in buying of clothes, shoes, books and pens for their school going children.

2.8: Benefits to the group

The group could able to collect the various NWFP products legally and sell the collected NWFPs. From the collection of annual membership fees and sale proceeds, the group can help individual member during misfortunes. They can also initiate other socioeconomic development activities and improve community vitality. Moreover, the group can also save the community

fund which they can use for sanctioning of members loan at low rate or they can be used for any livelihood support activities in the group.

PART 3: BY-LAWS-GROUP REGULATIONS AND NORMS

3.1: Members of the NWFP management & marketing group	Chiwog	Number of member households	Geog
	Gashari	9	Norbugang
	Total	9 HHs	
	Responsibilities and rights of members:		
	➤ The group members should ensure sustainable management of NWFP growing in the areas listed in the plan. ➤ The member should obey to the directives of the committee members. ➤ The members should abide to norms set in the plan for group governance and resources sharing. ➤ The members will sell NWFP products only through market channel set by the group. ➤ Every member of the group will have equal access to report to the committee members for violation of the management rules. ➤ The group members have right to reject the acceptance of re acceptance old member who has withdrawn and wanted to join again. Resign from the group owing to old age or no children to carry out work will be eligible to claim benefit from the total fund accumulated till the date of his/her resignation.		

3.2: Executive committee

3.2.1: Chairperson	Name of Chairperson: Mr. Phuntsho Wangdi
	Responsibilities of Coordinator/ Chairperson: ✓ To chair all kinds of meeting relating to NWFPs management and marketing. ✓ To resolves the disputes and conflict among the members. ✓ To enforce the fine and penalty. ✓ To explore market for the products. ✓ Fix date of collection and make an announcement to the group member. However, it will be done only after carrying out the rapid resource availability assessment conducted by the team. ✓ Coordinate annual marketing of NWFPs.

3.2.2: Secretary	Name of Secretary: Mr. Choni Dorji Responsibilities of Secretary: ✓ Provide assistance to the Chairman. ✓ Officiate Chairperson during his absence. ✓ Maintain records of the activities implemented, and also minutes of meeting. ✓ Write official correspondence. ✓ Compile the progress report and submitted to the GEFO for further submission to the Divisional Forest annually. ✓ Maintain NWFP collection and sale record.
3.2.3: Treasurer	Name of Treasurer: Mr. Sangay Dungjur Responsibilities of Treasurer: ➤ To maintain the financial transaction record and present the details report to Committee and the group members. ➤ Officiate as the Chairperson as Secretary in his absence. ➤ Maintain the cash transaction record submits expenditure and cash balance report to Chairman and present it to the group in meetings. ➤ Collection of membership fee from the group members. ➤ Collect fines and penalties as recommended by chairperson.
3.3: Compensation to the executive committee for marketing	During this plan period, the executive committee will work on voluntary basis and will not ask for any compensation.
3.4: Membership fee and purpose	Membership fee of Nu. 100/- annually from each members. If new member joins the group they should pay 150/-as membership registration fees and Nu.100 as annual membership fee. However, membership registration fee will be collected only once. The new member will have no right to claim benefit from the fund accumulated prior to registration as member. The collected fee will be deposited in the joint account of the group and will be used in work related to Community Development. The activity and ceiling of fund to be used will be approved in general meeting held every year.

3.5: Community development fund and utilization of fund	The bank account will be operated by Chairman, Secretary and Treasurer, and they are signatory authority to the group Bank Account. All expenditure incurred related to group activity will be met from the groups Account. Technical support will be sought from the nearest Forest Range Office The source of fund will be from collection of fine, membership fees and sales proceeds (5% of amount from sale of products) collected from the members. It will be collected irrespective of the quantity of products sold by the members. The fund accumulated will be equally shared/divided among the group members to once the bank account balance is enough (Nu.100,000) for division.
3.6: Conflict resolution	In case of disputes or conflicts arise among communities regarding NWFP harvesting and sell it will be discussed in the committee meeting. In case there is a conflict between the NWFP Group and outsiders, the case shall be forwarded to the Forest Range Office, Nganglam, and Gewog Administration by the coordinator. If the nature of the conflict is severe the case will be forwarded to the Gewog Tshogchung (GT) Dzongkhag and Tshogdue (DT) for further deliberation.

3.7: Penalties	In case, if the member of the NWFP Group violates the harvesting rules prescribed in the plan, he or she will not be allowed to harvest the NWFP for the rest of the season and the following season. Members collecting the NWFP prior to the fixed date/announcement made by the Coordinator, the offender will be levied fine Nu. 1000/- and product will be seized by the group. The seized products will be sold through auction and sale amount collected will be directly deposited in groups account. Individuals selling products illegally will be fine a sum if Nu.1000 per kg with seizer of products. Number of days for the collection will depend on availability of resources and general consensus by the members. Collection of NWFPs should be carried out by all household members. However, exemption will be made for those households which have only elderly people and problem related with misfortune. Only one member from each household will be entertained and additional collectors will be fine Nu. 500/- per head and the quantity collected will be surrendered to the group. Non-members are strictly not allowed and if apprehended fine of Nu. 500/- kg will be imposed with seizure of products. Absentee during meeting is liable to pay Nu. 200/- per head. An absentee during work is liable to pay Nu. 215/per head. If the members remain absent more than three times he/she will be expelled from the group. However, he/she will be warned in written for the possibility of expulsion from the group if remains absent once more. An absentee owing to death cases, sick or other genuine reasons are considered only after the approval from Chairperson. Remaining absent without approval from the chairperson will be liable fine Nu.200/-
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Table 4: Name list of Gashari NWFPs Management Group

Sl. No	Name	Chiwog	Designation	Signature or thump print
1	Mr. Phuntsho Wangdi	Gashari	Chairman	
2	Mr. Choni Dorji		Secretary	
3	Mr. Sangay Dungjur		Treasurer	
4	Mr. TENCHU		Member	
5	Mr. Chhimme		Member	
6	Mr. Kezang Wangchuk		Member	
7	Mr.Kezang B		Member	
8	Mr. Taw Gyeltshen		Member	
9	Mr. Tashi Dorji		Member	

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